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```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

df =
pd.read_csv("/workspaces/Mastery_in_DataAnalytics_and_Visualizations_a
nd_Career_Advancemen/WA_Fn-UseC_-HR-Employee-Attrition-
selectedcolumns.csv")

print(df.head())
print(df.shape)
```

```
   Age Attrition     BusinessTravel  DailyRate              Department

    41       Yes      Travel_Rarely       1102                   Sales

\
0

1   49        No  Travel_Frequently        279  Research & Development

2   37       Yes      Travel_Rarely       1373  Research & Development

3   33        No  Travel_Frequently       1392  Research & Development

4   27        No      Travel_Rarely        591  Research & Development


   DistanceFromHome  Education EducationField  EmployeeCount
EmployeeNumbe
                  1          2  Life Sciences              1 0
                                                             1
                  8          1  Life Sciences              1 1
                                                             2
                  2          2          Other              1 2
                                                             4
                  3          4  Life Sciences              1 3
                                                             5
                  2          1        Medical              1 4
                                                             7

(1470, 10)
```

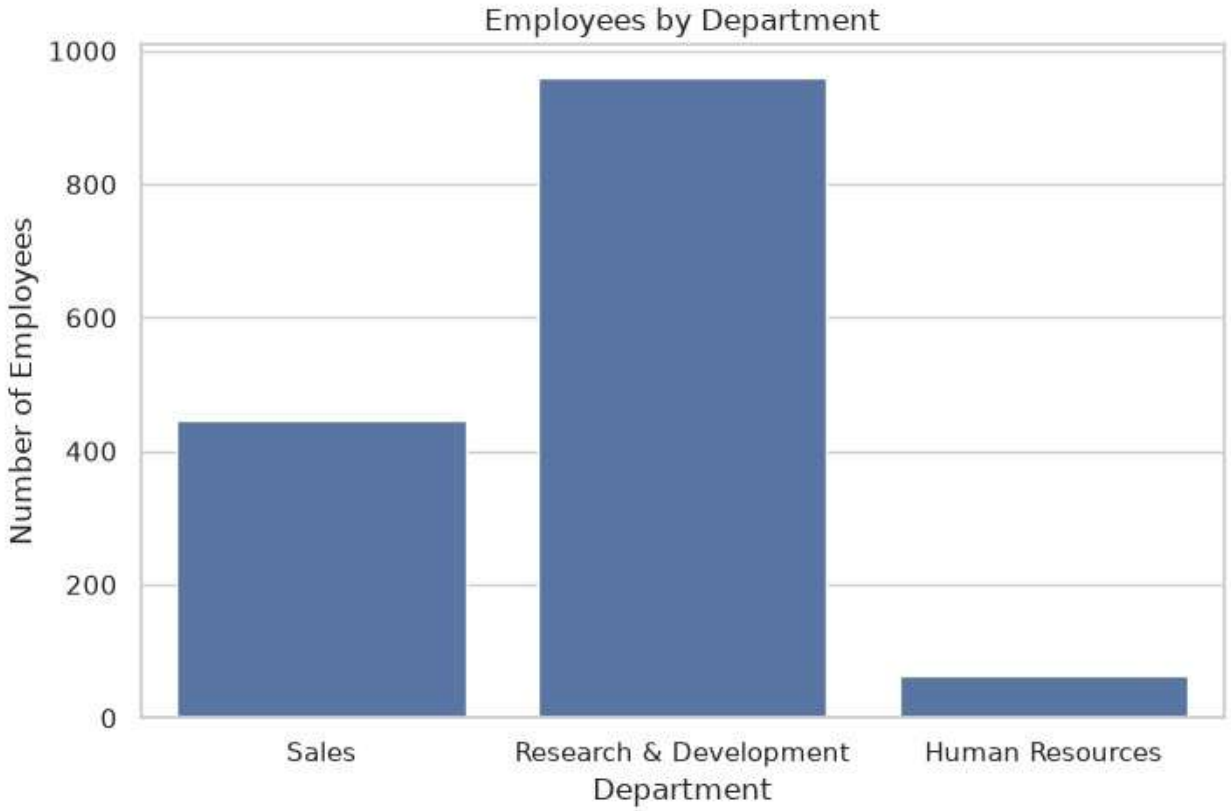
```
print(df.columns.tolist())
```

```
['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department',
'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount',
```

```
'EmployeeNumber']
plt.figure(figsize=(8, 5))
```

```
sns.countplot(data=df, x="Department")
```

```
plt.title("Employees by Department")
plt.xlabel("Department") plt.ylabel("Number of
Employees") plt.show()
```



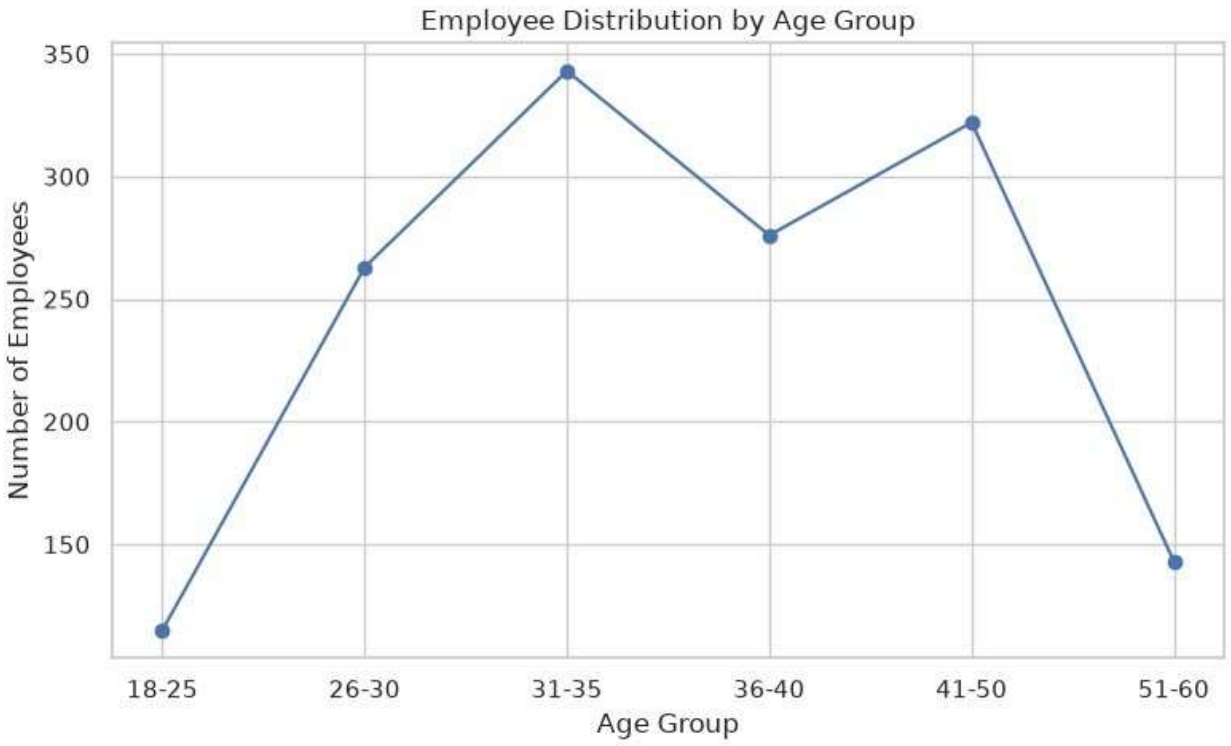

```
df["AgeGroup"] = pd.cut(
df["Age"],
    bins=[18, 25, 30, 35, 40, 50, 60],
    labels=["18-25", "26-30", "31-35", "36-40", "41-50", "51-60"]
)

age_count = df["AgeGroup"].value_counts().sort_index()

plt.figure(figsize=(9, 5))

plt.plot(age_count.index, age_count.values, marker="o")

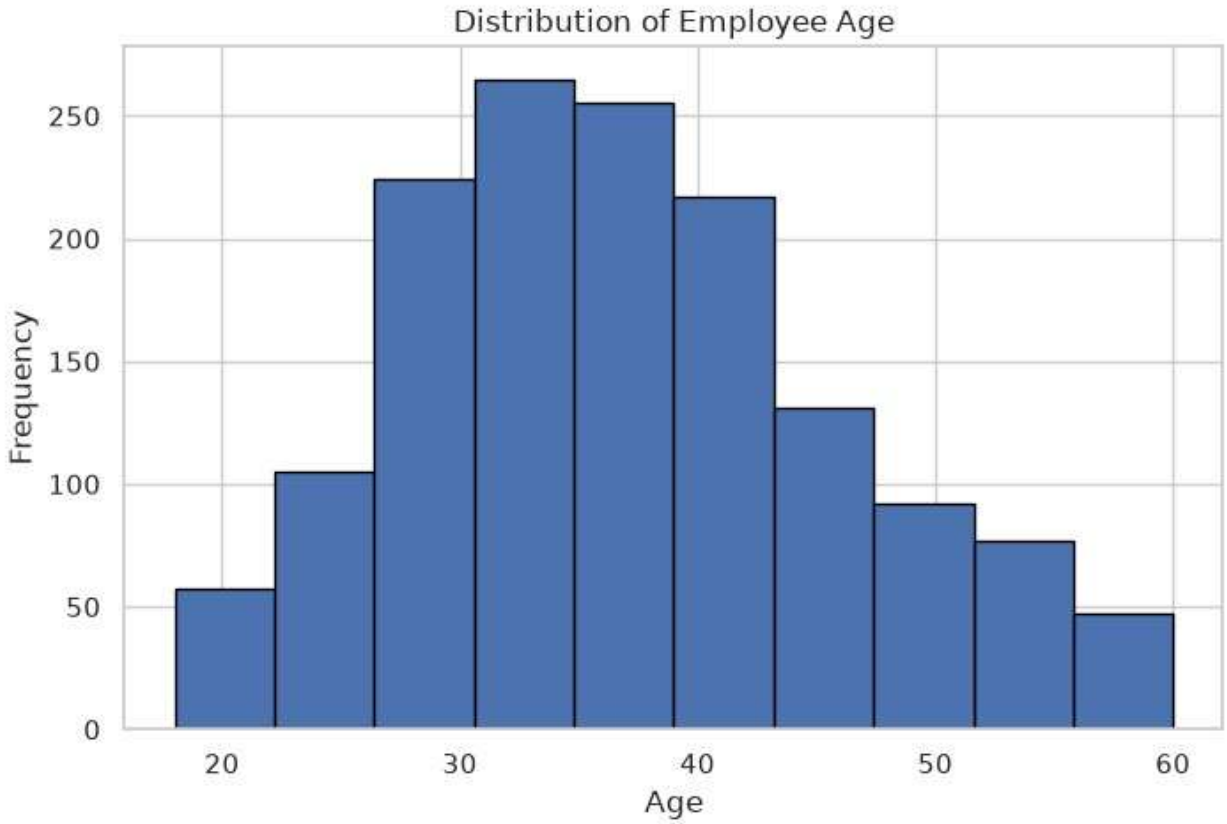
plt.title("Employee Distribution by Age Group")
plt.xlabel("Age Group") plt.ylabel("Number of
Employees")
plt.show()
```




```
plt.figure(figsize=(8, 5))

plt.hist(df["Age"], bins=10, edgecolor="black")

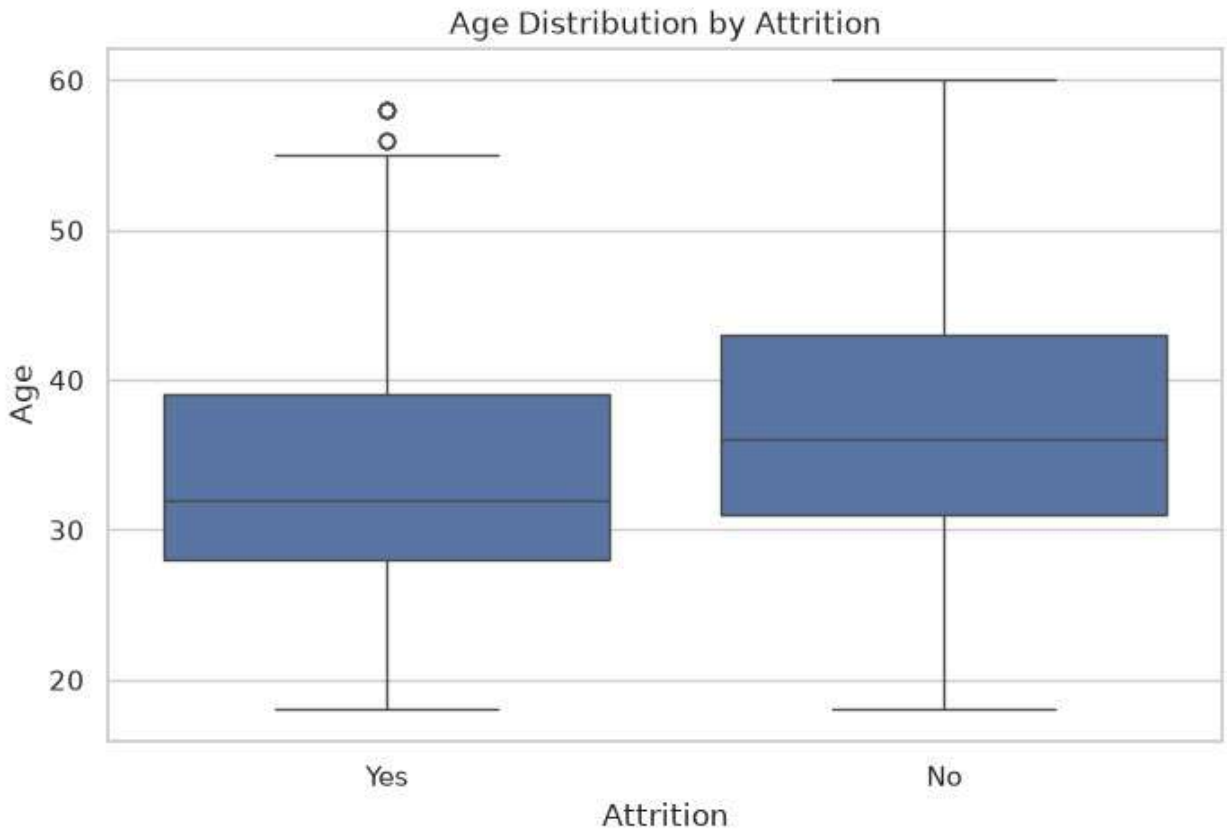
plt.title("Distribution of Employee Age")
plt.xlabel("Age") plt.ylabel("Frequency")
plt.show()
```




```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x="Attrition", y="Age")

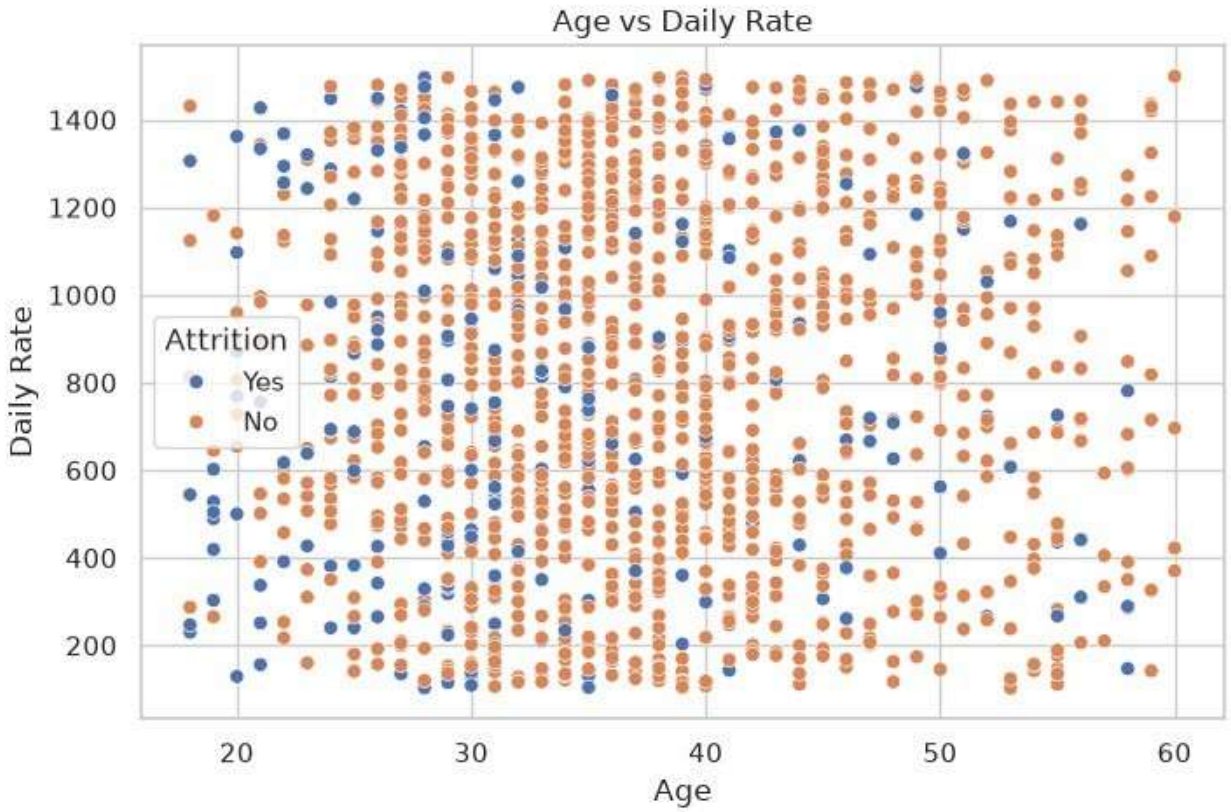
plt.title("Age Distribution by Attrition")
plt.xlabel("Attrition") plt.ylabel("Age")
plt.show()
```




```
plt.figure(figsize=(8, 5))

sns.scatterplot(
data=df,
x="Age",
y="DailyRate",
hue="Attrition"
)

plt.title("Age vs Daily Rate")
plt.xlabel("Age")
plt.ylabel("Daily Rate")
plt.show()
```

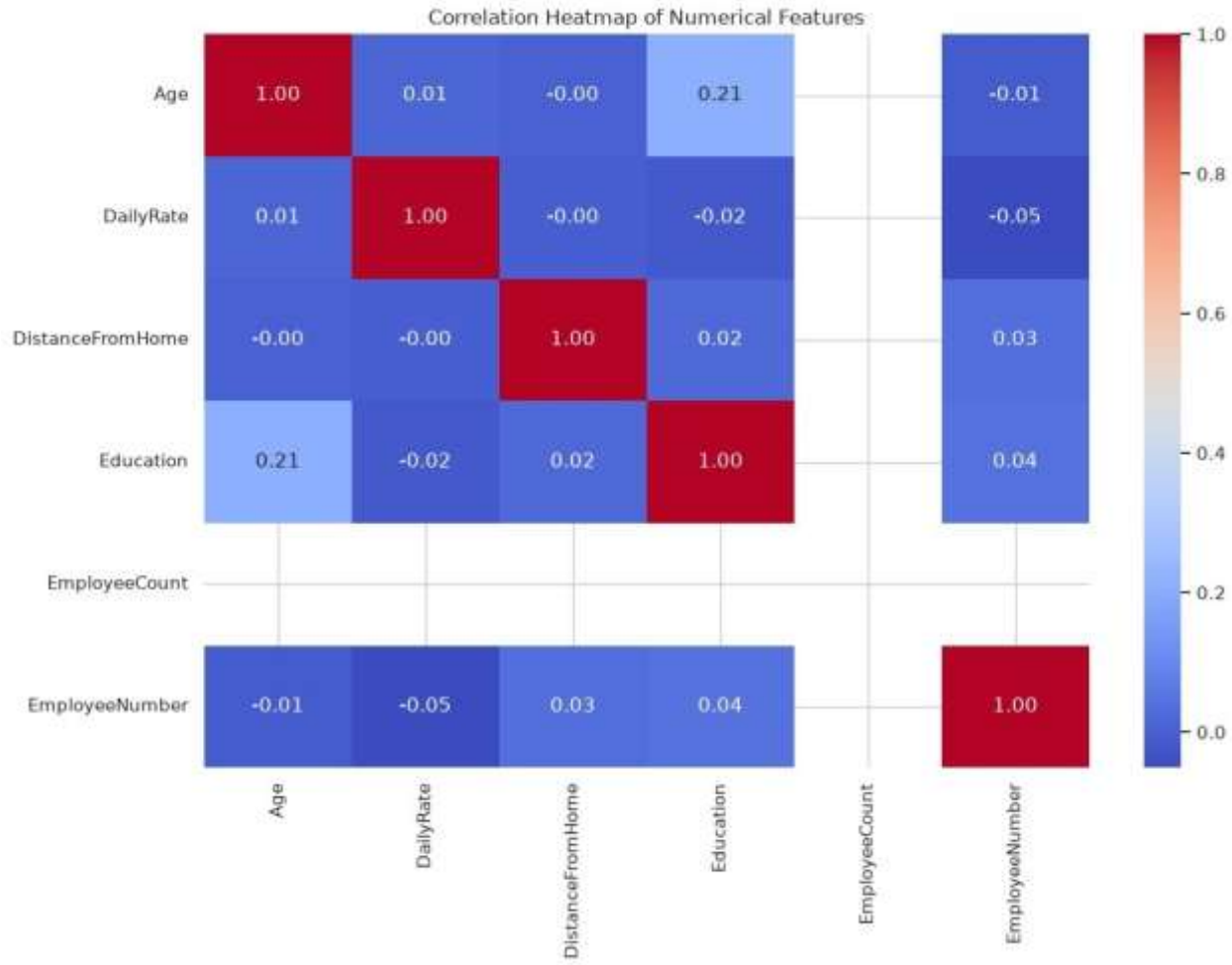



```
numeric_df = df.select_dtypes(include="number")

plt.figure(figsize=(12, 8))

sns.heatmap(
numeric_df.corr(),
annot=True,
cmap="coolwarm",
fmt=".2f"
)

plt.title("Correlation Heatmap of Numerical Features")
plt.show()
```

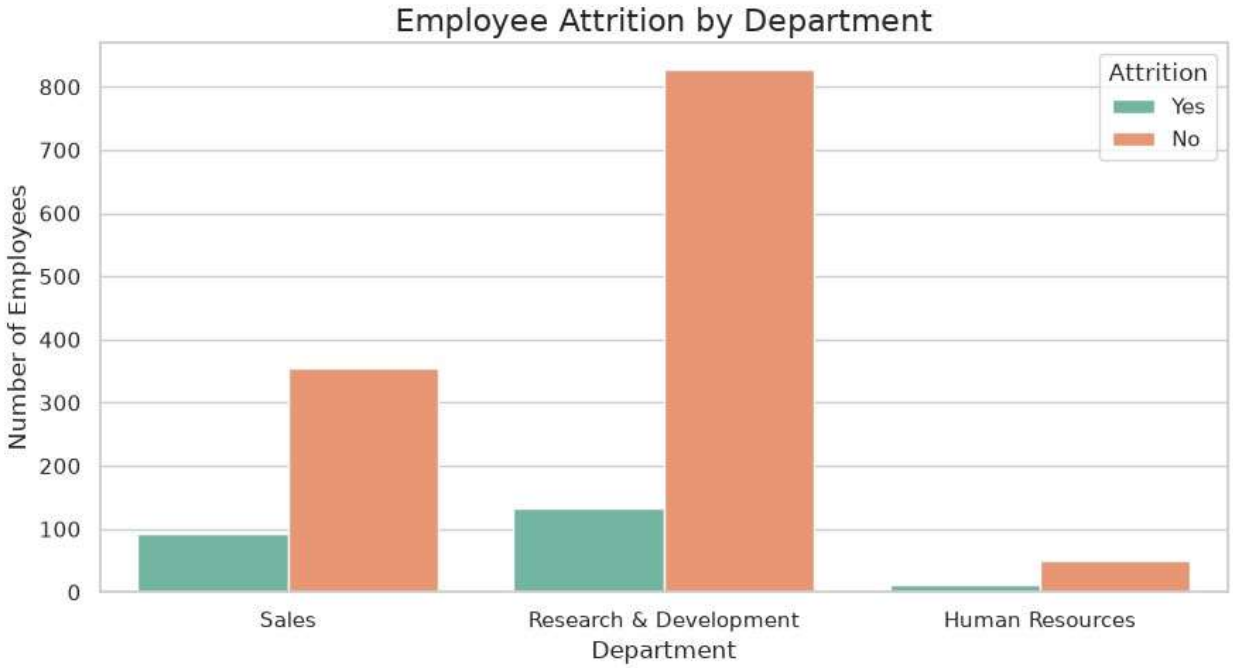



```
plt.figure(figsize=(9, 5))

sns.countplot(
data=df,
x="Department",
hue="Attrition",
palette="Set2"
)

plt.title("Employee Attrition by Department", fontsize=16)
plt.xlabel("Department", fontsize=12) plt.ylabel("Number
of Employees", fontsize=12) plt.legend(title="Attrition")

plt.tight_layout()
plt.show()
```


Employee Attrition by Department
Attrition
Yes
No
Number of Employees
800
700
600
500
400
300
200
100
0
Sales
Research & Development
Human Resources
Department

```
print("Total Employees:", len(df))

print("\nDepartment-wise Employees:")
print(df["Department"].value_counts())

print("\nAttrition Distribution:")
print(df["Attrition"].value_counts())
print("\nAverage Age:", df["Age"].mean())

print("\nAverage Daily Rate:", df["DailyRate"].mean())
```

```
Total Employees: 1470

Department-wise Employees:
Department
Research & Development    961
Sales                     446
Human Resources            63
Name: count, dtype: int64

Attrition Distribution:
Attrition
No     1233
Yes     237
Name: count, dtype: int64

Average Age: 36.923809523809524

Average Daily Rate: 802.4857142857143
```

QUESTIONS AND ANSWERS

1. Why is data visualization essential in data analytics?

Data visualization presents data through charts and graphs, making complex information easier to understand. It helps identify trends, patterns, comparisons, and outliers and supports better decision-making.

2. Difference between Matplotlib and Seaborn

- Matplotlib: General-purpose library with high customization.
- Seaborn: Built on Matplotlib and mainly designed for statistical visualization.

Seaborn is more suitable for statistical visualizations because it provides easy-to-use plots such as heatmaps, box plots, and regression plots.

3. Which chart is suitable for comparing sales across regions?

A bar chart is most suitable because regions are categories, and bars make it easy to compare their sales values and identify the highest and lowest sales.

4. What information do histogram and box plot provide?

- Histogram: Shows the distribution and frequency of numerical data.
- Box plot: Shows median, quartiles, spread, and outliers.

5. Purpose of a scatter plot

A scatter plot shows the relationship between two numerical variables. It helps identify positive, negative, or no correlation and can also reveal clusters and outliers.

6. What is a correlation heatmap?

A correlation heatmap displays correlation values between variables using colors. It helps identify strongly correlated features, which can assist in removing redundant features during machine learning feature selection.

7. Why is chart customization important?

Titles, labels, legends, and colors make charts clear and easy to understand. Proper customization helps the viewer interpret the data correctly.

8. Factors for selecting a visualization

Consider:

- Type of data
- Purpose of analysis
- Number of variables
- Audience
- Data size
- Readability

Examples: line chart for trends, bar chart for comparison, histogram for distribution, scatter plot for relationships.

9. How can misleading visualizations affect business decisions?

Misleading charts can cause incorrect interpretation and poor decisions. For example, using a truncated Y-axis can make a small increase in sales appear very large, causing management to overestimate business growth.

10. How does visualization support storytelling and business intelligence?

Visualization converts raw data into clear and meaningful insights. Dashboards and charts help organizations monitor KPIs, identify trends, understand problems, and make quick and informed business decisions.